

Mushood Hanif

Senior AI Engineer

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Summary

Senior AI Engineer with **nearly 7 years** of experience, grown from a junior machine learning role into leading agentic-systems architecture – across just two companies. Progressed from bilingual NLP classifiers, to production RAG and fraud-detection systems, to multi-agent LLM orchestration and on-prem fine-tuning infrastructure, with each stage building directly on the last. Focused on the layer most engineers skip: inference architecture, concurrency, cost per token, and failure recovery, not just prompt-and-ship. Comfortable owning a system from data pipeline through production serving through post-launch reliability.

Technical Skills

Languages: Python, TypeScript, SQL, Bash

AI/ML Frameworks: PyTorch, Hugging Face Transformers, LangChain, LangGraph, scikit-learn

LLM Engineering: RAG, LoRA/QLoRA, PEFT, Fine-tuning, Prompt Engineering, Agentic Systems, MCP, vLLM, Quantization (INT8/4-bit)

Backend & Infra: FastAPI, Celery, Redis, PostgreSQL, Docker, Kubernetes, CI/CD, Microservices, gRPC

Cloud: Azure (AKS, ML Studio), AWS (EC2, SageMaker), GCP (GKE, Vertex AI)

Search & Data: FAISS, HNSW, ChromaDB, Vector Search, ETL, Feature Stores, pandas, Apache Airflow

MLOps & Reliability: Observability, Load Testing, Canary Deployments, A/B Testing, Drift Monitoring, Blue-Green Deployment

Professional Experience

Senior AI Engineer

Sep 2025 – May 2026

Afiniti

Lahore, Pakistan

- Cut manual document-processing time from **3 days to under 4 hours (~92%)** for a logistics client, by architecting a 6-agent LangGraph orchestration layer spanning intake, extraction, and compliance review.
- Raised extraction accuracy from **81% to 96% F1**, by fine-tuning domain-specific LLaMA-3 models with QLoRA on a single on-prem A100 GPU to satisfy a strict data-residency requirement.
- Cut new-integration onboarding time from **~2 weeks to 2 days** across **9** internal APIs, by designing and building an MCP-based tool-routing layer adopted company-wide.
- Standardized evaluation practice across **3** other product teams, by building and leading rollout of a company-wide LLM eval harness now running **1,200+** automated test cases per release.
- Reduced GPU idle time by **35%** across the on-prem inference cluster, by implementing a dynamic batching and request-queuing scheduler.
- Cut agent hallucination rate from **11% to under 2%**, by introducing a citation-grounding and self-verification step into the agent pipeline.
- Grew a pod from **2 to 5** engineers in **10** months, by defining the team's technical roadmap and leading its hiring process.

AI Engineer

Jul 2023 – Aug 2025

Afiniti

Lahore, Pakistan

- Cut inference latency from **2.1s to under 180ms (~92%)** and per-transaction compute cost by **64%**, by migrating on-prem batch inference to a GPU-backed streaming service on Azure.
- Lowered false-positive fraud alerts from **14% to 3.5%** while raising catch-rate by **22%** across **1.2M+** daily transactions, by designing a real-time fraud-scoring pipeline.
- Cut production incident MTTR from **~50 to ~8 minutes (~84%)**, by introducing a canary-deployment and automated rollback framework.
- Increased model retraining frequency from monthly to weekly, by building an automated feature-drift detection and retraining-trigger system.
- Reduced false-decline rate on legitimate transactions by **9%**, by adding a secondary ensemble model validated against a **50,000+**-case labeled adjudication set.
- Cut onboarding time for new data scientists from **3 weeks to 5 days**, by authoring internal MLOps documentation and a self-serve feature store.

Machine Learning Engineer

Apr 2022 – Jun 2023

Confiz

Lahore, Pakistan

- Cut average customer ticket resolution time from **42 to 9 minutes (78%)** across **40,000+** monthly conversations, by designing and deploying a bilingual (Urdu/English) RAG support chatbot.
- Improved per-instance throughput by $\sim 8\times$ (**12→95 req/s**) while holding p95 latency under **400ms**, by migrating the request pipeline to async FastAPI + Celery.
- Cut manual QA-logging effort by **60% (~120 hrs/month)**, by implementing a FAISS + sentence-transformer retrieval layer with automated relevance logging.
- Increased chatbot containment rate (resolved without human handoff) from **48% to 71%**, by tuning a confidence-based escalation threshold.
- Cut average inference cost per conversation by **40%**, by quantizing the production embedding model from FP32 to INT8.

Junior Machine Learning Engineer

Nov 2019 – Mar 2022

Confiz

Lahore, Pakistan

- Reduced manual feedback-tagging effort by **70%** for the support team, by building a bilingual sentiment classifier integrated into the ticketing pipeline.
- Improved model retraining turnaround by **45%** across weekly cycles, by scripting a repeatable data-labeling and preprocessing pipeline in Python.
- Raised FAQ retrieval accuracy by **18 points (62%→80%)**, by introducing TF-IDF-based reranking ahead of the legacy keyword search.
- Cut labeling backlog from **3,000 to under 200** tickets in one quarter, by building a semi-automated active-learning labeling loop.
- Cut ramp-up time for **2** incoming interns by half, by documenting and training them on the team's data pipeline and evaluation practices.

Key Projects

Agentic Document Compliance Pipeline

Afiniti – Senior AI Engineer

- Delivered a production multi-agent system processing **1,000+** logistics documents weekly with **zero** unsupervised compliance failures, by combining LangGraph orchestration with a QLoRA-tuned extraction model and a human-in-the-loop review gate.
- Cut false-approval rate on ambiguous documents from **6% to under 1%**, by adding a confidence-scored routing layer that escalates low-certainty cases to human reviewers.
- Reduced average per-document processing cost by **55%**, by replacing a third-party OCR+LLM vendor call with an in-house fine-tuned extraction model.
- Cut root-cause debugging time for pipeline failures from **~2 hours to ~15 minutes**, by instrumenting full request tracing across all **6** agents.

On-Prem LLM Fine-Tuning & Serving Framework

Afiniti – Senior AI Engineer

- Cut average model-to-production time from **6 weeks to 9 days** across **4** internal teams, by building a standardized QLoRA fine-tuning and vLLM-based serving framework.
- Reduced GPU memory footprint per served model by **45%**, by implementing 4-bit quantization and adapter-swapping instead of full model duplication.
- Enabled **3** additional product teams to self-serve fine-tuned models without direct ML-engineering support, by building a CLI and internal documentation.

Real-Time Fraud Scoring Pipeline

Afiniti – AI Engineer

- Designed and shipped a streaming fraud-detection pipeline scoring **1.2M+** transactions/day at under **180ms** p95 latency, combining gradient-boosted models with a real-time feature store.
- Reduced false-positive rate from **14% to 3.5%** while raising fraud catch-rate by **22%**, by layering a secondary ensemble model behind the primary classifier.
- Cut model-drift-related incidents by **80%** over **6** months, by building automated feature-drift monitoring with alerting thresholds.

Azure GPU-Backed Streaming Inference Migration

Afiniti – AI Engineer

- Cut inference latency from **2.1s to under 180ms (~92%)**, by migrating a legacy on-prem batch-inference system to a GPU-backed streaming service on Azure.

- Reduced per-transaction compute cost by **64%**, by right-sizing GPU instance types and introducing autoscaling based on real-time queue depth.
- Cut deployment rollback time from **~40 minutes to under 5 minutes**, by containerizing the inference service and introducing blue-green deployment.

Bilingual RAG Support Chatbot

Confiz – Machine Learning Engineer

- Cut average resolution time from **42 to 9 minutes (78%)** across **40,000+** monthly conversations, by designing and deploying a bilingual (Urdu/English) RAG support chatbot.
- Improved per-instance throughput **~8× (12→95 req/s)** while holding p95 latency under **400ms**, by migrating the request pipeline to async FastAPI + Celery.
- Raised containment rate (no human handoff needed) from **48% to 71%**, by tuning a confidence-based escalation threshold.

FAISS-Based FAQ Retrieval Engine

Confiz – Machine Learning Engineer

- Cut hallucinated answers by **~80%**, by rebuilding the retrieval layer to serve **95%** of FAQ queries from a vetted knowledge base instead of live generation.
- Reduced average retrieval latency from **340ms to 60ms**, by replacing a brute-force cosine-similarity search with a FAISS HNSW index.
- Cut manual QA-logging effort by **60% (~120 hrs/month)**, by adding automated relevance logging on every retrieval call.

Urdu-English Sentiment Classifier

Confiz – Junior Machine Learning Engineer

- Reached **89%** accuracy on live customer feedback, by fine-tuning a lightweight BERT variant on **12,000** manually labeled support tickets – first model taken end-to-end from data labeling to production.
- Reduced manual feedback-tagging effort by **70%** for the support team, by integrating the classifier directly into the ticketing pipeline.
- Cut model retraining turnaround by **45%**, by scripting a repeatable data-labeling and preprocessing pipeline.

FAQ Search Reranking Prototype

Confiz – Junior Machine Learning Engineer

- Raised FAQ retrieval accuracy by **18 points (62%→80%)**, by introducing TF-IDF-based reranking ahead of the legacy keyword search – first production model shipped independently.
- Cut labeling backlog from **3,000 to under 200** tickets in one quarter, by building a semi-automated active-learning labeling loop later adopted by the wider team.

Education

BS, Software Engineering (Hons.)

Oct 2014 – Aug 2019

GIFT University, Gujranwala, Pakistan

- Vice President, Drama Society
- Technical Head, YCPS (Young Computer Professionals Society)
- Best Commercial Capstone Award
- CGPA: 3.0/4.0